

Oral Microbiome Freezer Stability Study: Manuscript End-to-End Tables and Figures

1. Input Data

This report starts from the analysis table and generated result files in the current working directory:

```
if (!exists("params")) {
  params <- list(run_icc_calculation = TRUE, run_adnois = TRUE, icc_iterations = 100L)
}
run_icc_calculation <- isTRUE(params$run_icc_calculation)
run_adnois <- isTRUE(params$run_adnois)
icc_iterations <- as.integer(params$icc_iterations)

analysis_table_file <- "analysis_sample_table_304.csv"
icc_buccal_file <- "result_ICC_Storage_Time_KingFisher_Buccal_Cells_-80C.txt"
icc_saliva_file <- "result_ICC_Storage_Time_KingFisher_Saliva_-80C.txt"
adonis2_file <- "result_adonis2.csv"

distance_files <- c(
  "Bray Curtis" = "distance_bray_curtis_study_235.txt",
  "Jaccard" = "distance_jaccard_study_235.txt",
  "Unweighted Unifrac" = "distance_unweighted_unifrac_study_235.txt",
  "Generalized Unifrac" = "distance_generalized_unifrac_study_235.txt",
  "Weighted Unifrac" = "distance_weighted_unifrac_study_235.txt"
)

input_files_table <- data.frame(
  check.names = FALSE,
  "Input file" = c(
    analysis_table_file,
    icc_buccal_file,
    icc_saliva_file,
    adonis2_file,
    unname(distance_files)
  ),
  "Used for" = c(
    "Analysis sample metadata, alpha diversity, PCoA coordinates, relative abundance, CLR values",
    "Buccal cell ICC and 95% CI table/figures",
    "Saliva ICC and 95% CI table/figures",
    "PERMANOVA / Adonis2 table and barplot",
    "Bray Curtis distance matrix",
    "Jaccard distance matrix",
    "Unweighted UniFrac distance matrix",
    "Generalized UniFrac distance matrix",
    "Weighted UniFrac distance matrix"
  ),
  "Required when" = c(
    "Always",
    "run_icc_calculation = FALSE",
    "run_icc_calculation = FALSE",
    "run_adnois = FALSE",
  )
)
```

```

    rep("run_adnois = TRUE", length(distance_files))
  )
)

required_inputs <- analysis_table_file
if (run_icc_calculation) {
  required_inputs <- c(required_inputs)
} else {
  required_inputs <- c(required_inputs, icc_buccal_file, icc_saliva_file)
}

if (run_adnois) {
  required_inputs <- c(required_inputs, distance_files)
} else {
  required_inputs <- c(required_inputs, adonis2_file)
}

missing_inputs <- required_inputs[!file.exists(required_inputs)]
if (length(missing_inputs) > 0L) {
  stop("Missing required input file(s): ", paste(missing_inputs, collapse = ", "))
}

```

```

library(dplyr)
library(forcats)
library(ggplot2)
library(knitr)
library(patchwork)
library(tidyr)
library(vegan)

latex_escape <- function(x) {
  x <- gsub("\\\\", "\\\textbackslash{}", x)
  x <- gsub("([#$%&_{}])", "\\\textasciitilde{}", x, perl = TRUE)
  x <- gsub("~", "\\\textasciitilde{}", x, fixed = TRUE)
  x <- gsub("^", "\\\textasciicircum{}", x, fixed = TRUE)
  x
}

format_table <- function(x, caption, digits = 8, wide = FALSE, row_names = NA) {
  if (knitr::is_latex_output()) {
    kbl <- knitr::kable(
      x,
      format = "latex",
      digits = digits,
      booktabs = FALSE,
      row.names = row_names,
      format.args = list(scientific = FALSE)
    )
    table_body <- if (wide) {
      paste0("\\resizebox{\\linewidth}{!}{%\\n", kbl, "\\n}")
    } else {
      kbl
    }
    font_size <- if (wide) "\\scriptsize" else "\\small"
    knitr::asis_output(paste0(
      "\\noindent\\textbf{", latex_escape(caption), "\\par\\vspace{0.35em}\\n",
      "\\begin{group}\\n", font_size, "\\n",
      table_body,

```

```

    "\n\\endgroup\\par\\vspace{0.8em}\\n"
  ))
} else {
  knitr::kable(
    x,
    digits = digits,
    caption = caption,
    row.names = row_names,
    format.args = list(scientific = FALSE)
  )
}
}

```

```

format_table(
  input_files_table,
  caption = "Input files used by the report.",
  wide = TRUE,
  row_names = FALSE
)

```

Input files used by the report.

Input file	Used for	Required when
analysis_sample_table_304.csv	Analysis sample metadata, alpha diversity, PCoA coordinates, relative abundance, CLR values	Always
result_ICC_Storage_Time_KingFisher_Buccal_Cells_-80C.txt	Buccal cell ICC and 95% CI table/figures	run_icc_calculation = FALSE
result_ICC_Storage_Time_KingFisher_Saliva_-80C.txt	Saliva ICC and 95% CI table/figures	run_icc_calculation = FALSE
result_adonis2.csv	PERMANOVA / Adonis2 table and barplot	run_adonis = FALSE
distance_bray_curtis_study_235.txt	Bray Curtis distance matrix	run_adonis = TRUE
distance_jaccard_study_235.txt	Jaccard distance matrix	run_adonis = TRUE
distance_unweighted_unifrac_study_235.txt	Unweighted UniFrac distance matrix	run_adonis = TRUE
distance_generalized_unifrac_study_235.txt	Generalized UniFrac distance matrix	run_adonis = TRUE
distance_weighted_unifrac_study_235.txt	Weighted UniFrac distance matrix	run_adonis = TRUE

The table below records the sample counts and run settings used by this report.

```

analysis_table <- read.csv(
  analysis_table_file,
  check.names = FALSE,
  stringsAsFactors = FALSE,
  na.strings = "NA"
)

metadata_cols <- c(
  "Sample_ID",
  "Sample_Type",
  "Subject_ID",
  "Total_Reads",
  "Specimen_Storage_Time",
  "DNA_Storage_Time",
  "Extraction_Method",
  "Source_Material",
  "Specimen_Storage_Temp"
)

alpha_cols <- c("observed_features", "shannon_entropy", "pielou_evenness", "faith_pd")

missing_cols <- setdiff(c(metadata_cols, alpha_cols), colnames(analysis_table))
if (length(missing_cols) > 0L) {
  stop("Missing expected analysis table column(s): ", paste(missing_cols, collapse = ", "))
}

sample_type_order <- c(
  "artificialcolony",
  "ExtractionBlank",

```

```

"PCRNTCBlank",
"PCRWaterBlank",
"robogut",
"Study"
)

study_data <- analysis_table[analysis_table$Sample_Type == "Study", , drop = FALSE]
study_data$Specimen_Storage_Time <- factor(
  study_data$Specimen_Storage_Time,
  levels = c("Baseline", "Month1", "Month12", "Year5")
)

masked_subject_order <- c(
  "TV 0012", "TV 0026", "TV 0027", "TV 0024", "TV 0023",
  "TV 0009", "TV 0025", "TV 0013", "TV 0005", "TV 0002",
  "TV 0021", "TV 0010", "TV 0016", "TV 0022", "TV 0006",
  "TV 0007", "TV 0018", "TV 0019", "TV 0017", "TV 0011",
  "TV 0001", "TV 0028", "TV 0004", "TV 0030", "TV 0003",
  "TV 0014", "TV 0029", "TV 0015", "TV 0020", "TV 0008"
)
if (all(masked_subject_order %in% study_data$Subject_ID) &&
    length(unique(study_data$Subject_ID)) == length(masked_subject_order)) {
  study_data$Subject_ID <- factor(study_data$Subject_ID, levels = masked_subject_order)
} else {
  study_data$Subject_ID <- factor(study_data$Subject_ID, levels = sort(unique(study_data$Subject_ID)))
}
study_data$Participant_ID <- sub("^TV[ _]+", "", study_data$Subject_ID)
rownames(study_data) <- study_data$Sample_ID

time_levels_raw <- c("Baseline", "Month1", "Month12", "Year5")
time_levels_plot <- c("Baseline", "1 Month", "12 Months", "5 Years")
time_labels <- stats::setNames(time_levels_plot, time_levels_raw)
study_data$Specimen_Storage_Time_Plot <- factor(
  time_labels[as.character(study_data$Specimen_Storage_Time)],
  levels = time_levels_plot
)

outcome_vars <- setdiff(colnames(study_data), c(metadata_cols, "Participant_ID", "Specimen_Storage_Time_Plot"))

stopifnot(nrow(analysis_table) == 304L)
stopifnot(nrow(study_data) == 235L)

input_summary <- data.frame(
  Item = c(
    "Total analysis samples",
    "QC samples",
    "Study samples",
    "Run ICC calculation",
    "Run Adonis2 calculation",
    "ICC iterations"
  ),
  Value = c(
    nrow(analysis_table),
    sum(analysis_table$Sample_Type != "Study"),
    nrow(study_data),
    run_icc_calculation,
    run_adonis,
    icc_iterations
  )
)

```

```

),
  check.names = FALSE
)

format_table(input_summary, caption = "Input data and run settings.")

```

Input data and run settings.

Item	Value
Total analysis samples	304
QC samples	69
Study samples	235
Run ICC calculation	1
Run Adonis2 calculation	1
ICC iterations	100

```

summary_stats <- function(x) {
  x <- as.numeric(x)
  x <- x[!is.na(x)]
  if (length(x) == 0L) {
    return(c(n = 0, min = NA, mean = NA, median = NA, max = NA, sd = NA, cv = NA))
  }

  x_mean <- mean(x)
  x_sd <- if (length(x) > 1L) stats::sd(x) else NA_real_
  c(
    n = length(x),
    min = min(x),
    mean = x_mean,
    median = stats::median(x),
    max = max(x),
    sd = x_sd,
    cv = if (!is.na(x_sd) && x_mean != 0) x_sd / x_mean else NA_real_
  )
}

summary_by_group <- function(dat, value_col, group_col, group_values, group_label) {
  rows <- lapply(group_values, function(group_value) {
    idx <- dat[[group_col]] == group_value
    stats <- summary_stats(dat[[value_col]][idx])
    data.frame(
      check.names = FALSE,
      Group = group_value,
      t(stats)
    )
  })
  out <- do.call(rbind, rows)
  names(out)[1] <- group_label
  out
}

alpha_summary_by_group <- function(dat, group_col, group_values, group_label) {
  rows <- lapply(group_values, function(group_value) {
    idx <- dat[[group_col]] == group_value
    group_dat <- dat[idx, , drop = FALSE]
    metric_summary <- unlist(lapply(alpha_cols, function(metric) {
      stats <- summary_stats(group_dat[[metric]])
      c(mean = stats[["mean"]], sd = stats[["sd"]], cv = stats[["cv"]])
    }

```

```

}))
names(metric_summary) <- as.vector(t(outer(alpha_cols, c("mean", "sd", "cv"), paste, sep = "_")))

data.frame(
  check.names = FALSE,
  Group = group_value,
  N_after_rarefaction_20000 = sum(!is.na(group_dat$observed_features)),
  t(metric_summary)
)
})

out <- do.call(rbind, rows)
names(out)[1] <- group_label
out
}

```

2. Reads Depth Summary Table

The read-depth summary uses the 69 QC samples before rarefaction and the 235 study samples after rarefaction, exactly as represented in the analysis table.

```

read_depth_summary <- summary_by_group(
  analysis_table,
  value_col = "Total_Reads",
  group_col = "Sample_Type",
  group_values = sample_type_order,
  group_label = "Sample_Type"
)

format_table(
  read_depth_summary,
  digits = 8,
  caption = "Read-depth summary for QC samples before rarefaction and study samples after rarefaction."
)

```

Read-depth summary for QC samples before rarefaction and study samples after rarefaction.

Sample_Type	n	min	mean	median	max	sd	cv
artificialcolony	17	24581	41178.764706	35045	83280	17316.718398	0.4205255
ExtractionBlank	15	0	9581.133333	364	83256	23816.811487	2.4858032
PCRNTCBlank	13	0	1.923077	0	9	3.402488	1.7692936
PCRWaterBlank	13	0	5.461538	6	20	6.063764	1.1102666
robogut	11	25660	42720.636364	34415	82233	18840.632401	0.4410195
Study	235	20293	56722.714894	58094	101179	20102.622650	0.3544016

3. Alpha Diversity Summary Table

The alpha-diversity summary reports observed ASVs, Shannon Index, Pielou's evenness, and Faith's phylogenetic diversity.

```

artificial_subject_order <- sort(unique(analysis_table$Subject_ID[analysis_table$Sample_Type == "artificialcolony"]))

alpha_summary_sample_type <- alpha_summary_by_group(
  analysis_table,
  group_col = "Sample_Type",
  group_values = sample_type_order,
  group_label = "Sample_Type"
)

```

```
)

alpha_summary_artificialcolony <- alpha_summary_by_group(
  analysis_table[analysis_table$Sample_Type == "artificialcolony", , drop = FALSE],
  group_col = "Subject_ID",
  group_values = artificial_subject_order,
  group_label = "Subject_ID"
)

format_table(alpha_summary_sample_type, digits = 8, caption = "Alpha diversity summary by sample type.", wide = 100)
```

Alpha diversity summary by sample type.

Sample_Type	N after rarefaction 20000	observed_features_mean	observed_features_sd	observed_features_cv	shannon_entropy_mean	shannon_entropy_sd	shannon_entropy_cv	pielou_evenness_mean	pielou_evenness_sd	pielou_evenness_cv	faith_pd_mean	faith_pd_sd	faith_pd_cv
artificialcolony	19	27.47059	10.585504	0.38570357	1.591766	0.73526971	0.20470980	0.7650059	0.09911022	0.12955489	4.096299	1.433544	0.3025198
ExtractionBlank	2	145.56000	9.192388	0.06317793	5.303199	0.06438828	0.01214140	0.7382566	0.00041011	0.00055552	11.666329	0.8732257	0.07485069
PCRBlank	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
PCRWaterBlank	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
robogut	11	82.00000	11.198214	0.13656139	3.581698	0.31143853	0.08695834	0.5648848	0.05019232	0.08885408	6.168312	0.9019369	0.16081172
Study	235	130.40426	27.662973	0.21136483	4.899211	0.39250323	0.08031196	0.7056995	0.03431162	0.05201264	10.385319	1.495913	0.14217968

```
format_table(alpha_summary_artificialcolony, digits = 8, caption = "Alpha diversity summary for artificial colony samples by subject ID.", wide = 100)
```

Alpha diversity summary for artificial colony samples by subject ID.

Subject_ID	N after rarefaction 20000	observed_features_mean	observed_features_sd	observed_features_cv	shannon_entropy_mean	shannon_entropy_sd	shannon_entropy_cv	pielou_evenness_mean	pielou_evenness_sd	pielou_evenness_cv	faith_pd_mean	faith_pd_sd	faith_pd_cv
D6300	1	12.00000	NA	NA	2.7500771	NA	NA	0.5331975	NA	NA	2.135883	NA	NA
D6310	1	5.00000	NA	NA	0.9686062	NA	NA	0.4171560	NA	NA	1.062212	NA	NA
D6331	1	20.00000	NA	NA	3.4624759	NA	NA	0.8474171	NA	NA	3.312291	NA	NA
DZ35316	11	29.45455	4.321195	0.1467072	3.8339719	0.07947437	0.02073658	0.7885014	0.03595278	0.04557323	5.216706	0.6459978	0.1238325
DZ35323	1	25.00000	NA	NA	3.8712437	NA	NA	0.8336270	NA	NA	4.431079	NA	NA
DZ35327	1	26.00000	NA	NA	3.6147069	NA	NA	0.7817581	NA	NA	4.707839	NA	NA
RSM0001	1	55.00000	NA	NA	4.0124641	NA	NA	0.6940347	NA	NA	6.754023	NA	NA

4. Supplemental Figure 1

Supplemental Figure 1 shows the first two PCoA axes for the five beta-diversity distances among analysis samples.

```
pcoa_rows <- analysis_table[!is.na(analysis_table$bray_curtis_PC1), , drop = FALSE]

sample_type_colors <- c(
  artificialcolony = "red",
  ExtractionBlank = "blue",
  robogut = "green",
  Study = grDevices::rgb(0.7, 0.7, 0.7, 0.5)
)

point_colors <- unname(sample_type_colors[pcoa_rows$Sample_Type])
point_colors[is.na(point_colors)] <- "black"

point_pch <- rep(20, nrow(pcoa_rows))
point_pch[pcoa_rows$Subject_ID == "D6300"] <- 1
point_pch[pcoa_rows$Subject_ID == "D6310"] <- 2
point_pch[pcoa_rows$Subject_ID == "D6331"] <- 3
point_pch[pcoa_rows$Subject_ID == "DZ35323"] <- 4
point_pch[pcoa_rows$Subject_ID == "DZ35327"] <- 5
point_pch[pcoa_rows$Subject_ID == "RSM0001"] <- 6

legend_text <- c(
  paste("Artificial community", c("DZ35316", "D6300", "D6310", "D6331", "DZ35323", "DZ35327", "RSM0001")),
  "ExtractionBlank",
  "robogut",
  "Study"
)

legend_colors <- c(rep("red", 7), "blue", "green", grDevices::rgb(0.7, 0.7, 0.7, 0.5))
legend_pch <- c(20, 1:6, 20, 20, 20)
```

```

plot_pcoa <- function(prefix, main, legend_position) {
  plot_data <- as.matrix(pcoa_rows[, paste0(prefix, c("_PC1", "_PC2"))])
  colnames(plot_data) <- c("PC1", "PC2")
  plot(plot_data, col = point_colors, pch = point_pch, main = main)
  legend(legend_position, legend_text, col = legend_colors, pch = legend_pch, cex = 1.2)
}

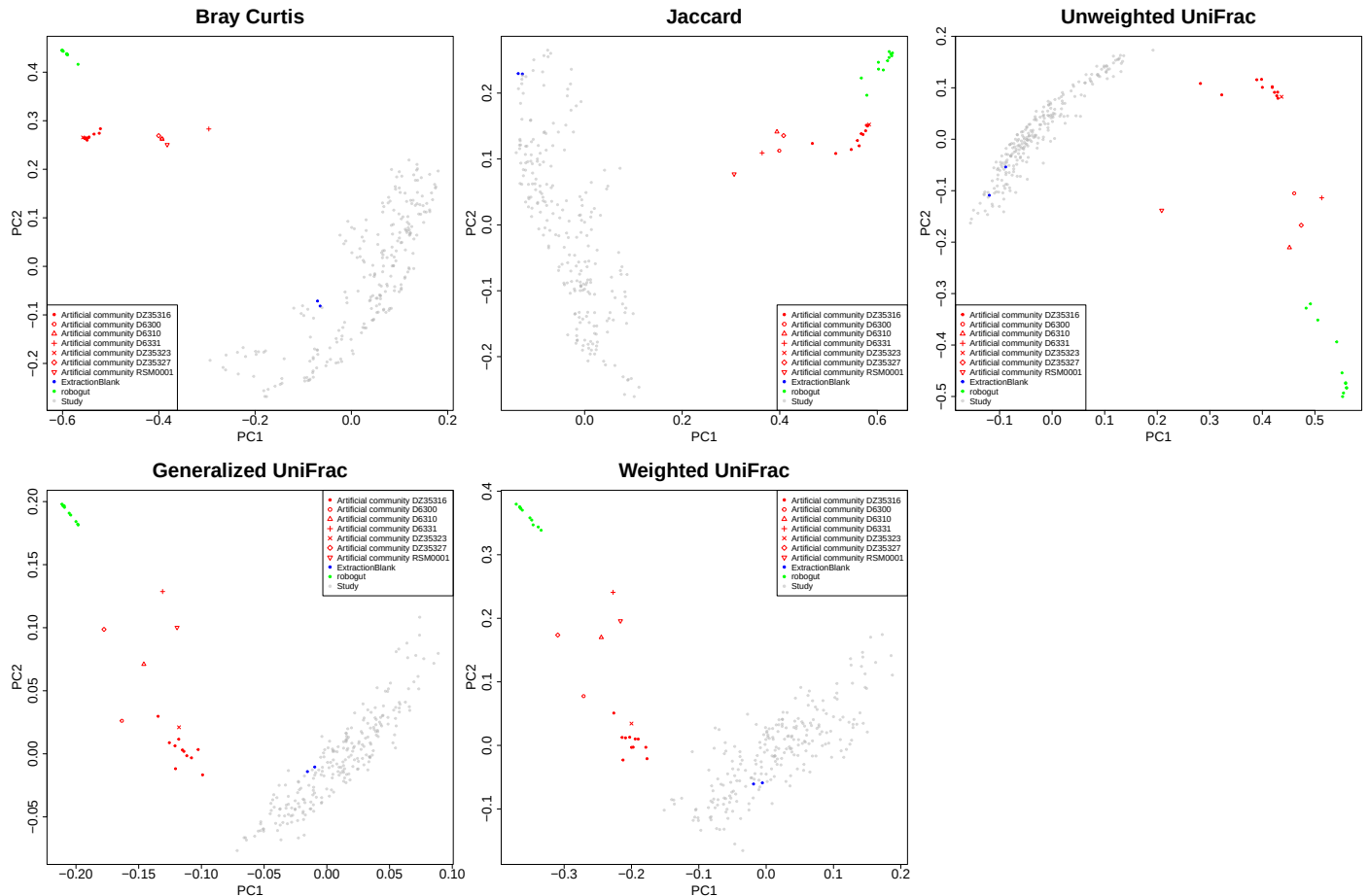
draw_supplemental_figure_1 <- function() {
  old_par <- par(mfrow = c(2, 3), cex.main = 3, cex.lab = 2, cex.axis = 2, mar = c(5, 5, 5, 1))
  on.exit(par(old_par), add = TRUE)
  plot_pcoa("bray_curtis", "Bray Curtis", "bottomleft")
  plot_pcoa("jaccard", "Jaccard", "bottomright")
  plot_pcoa("unweighted_unifrac", "Unweighted UniFrac", "bottomleft")
  plot_pcoa("generalized_unifrac", "Generalized UniFrac", "topright")
  plot_pcoa("weighted_unifrac", "Weighted UniFrac", "topright")
}

pdf("Supplemental Figure 1.pdf", width = 22.5, height = 15)
draw_supplemental_figure_1()
invisible(dev.off())

png("Supplemental Figure 1.png", width = 6750, height = 4500, res = 300)
draw_supplemental_figure_1()
invisible(dev.off())

knitr::include_graphics("Supplemental Figure 1.png")

```



Supplemental Figure 1: Principal coordinates analysis plots of the study samples and quality controls for the 5 beta

diversity metrics. The points are colored by type to indicate whether it is a study sample or quality control. For the quality controls, the artificial communities are further identified by shape to indicate the different communities.

5. Relative Abundance Table

This table lists the 14 selected genera with average relative abundance at least 0.01 in either buccal or saliva samples.

```
selected_genera_order <- c(
  "Prevotella",
  "Streptococcus",
  "Haemophilus",
  "Veillonella",
  "Neisseria",
  "Alloprevotella",
  "Rothia",
  "Fusobacterium",
  "Gemella",
  "Leptotrichia",
  "Granulicatella",
  "Actinomyces",
  "Porphyromonas",
  "Megasphaera"
)

selected_ra_cols <- paste0("RA_", selected_genera_order)
missing_ra_cols <- setdiff(selected_ra_cols, colnames(study_data))
if (length(missing_ra_cols) > 0L) {
  stop("Missing selected genus RA column(s): ", paste(missing_ra_cols, collapse = ", "))
}

buccal_data <- study_data[study_data$Source_Material == "Buccal Cells", , drop = FALSE]
saliva_data <- study_data[study_data$Source_Material == "Saliva", , drop = FALSE]

average_ra_table <- data.frame(
  check.names = FALSE,
  Genus = selected_genera_order,
  Buccal_and_Saliva = vapply(selected_ra_cols, function(col) mean(study_data[[col]], na.rm = TRUE), numeric(1)),
  Buccal = vapply(selected_ra_cols, function(col) mean(buccal_data[[col]], na.rm = TRUE), numeric(1)),
  Saliva = vapply(selected_ra_cols, function(col) mean(saliva_data[[col]], na.rm = TRUE), numeric(1))
)
colnames(average_ra_table) <- c(
  "Genus",
  sprintf("Buccal and Saliva (n = %d)", nrow(study_data)),
  sprintf("Buccal (n = %d)", nrow(buccal_data)),
  sprintf("Saliva (n = %d)", nrow(saliva_data))
)
rownames(average_ra_table) <- NULL

format_table(
  average_ra_table,
  digits = 10,
  caption = "Selected genera average relative abundance.",
  row_names = FALSE
)
```

Selected genera average relative abundance.

Genus	Buccal and Saliva (n = 235)	Buccal (n = 119)	Saliva (n = 116)
Prevotella	0.259618888	0.218072187	0.302240072
Streptococcus	0.233256786	0.297698401	0.167148577
Haemophilus	0.083804618	0.072912689	0.094978235
Veillonella	0.067496657	0.061421072	0.073729370
Neisseria	0.062867056	0.054855414	0.071085895
Alloprevotella	0.047738083	0.044126086	0.051443493
Rothia	0.037240406	0.051436944	0.022676715
Fusobacterium	0.035994691	0.032019143	0.040073055
Gemella	0.023099510	0.036603296	0.009246488
Leptotrichia	0.016799453	0.013104681	0.020589779
Granulicatella	0.015411776	0.015255965	0.015571616
Actinomyces	0.008538305	0.013695137	0.003248107
Porphyromonas	0.010178720	0.008546209	0.011853452
Megasphaera	0.008924010	0.006737797	0.011166762

5.1 Figure 1

Figure 1 summarizes mean genus-level relative abundance across specimen storage time and source material.

```

ra_cols <- grep("^RA_", colnames(study_data), value = TRUE)
ra_cols <- setdiff(ra_cols, grep("^RA_clr_", colnames(study_data), value = TRUE))
taxa_include <- sub("^RA_", "", names(sort(colMeans(study_data[, ra_cols], na.rm = TRUE), decreasing = TRUE)))

genus_colors <- c(
  "#34b0b5", "#efb586", "#e3e863", "#acd84e", "#9edbef",
  "#b7a6ea", "#5fbc21", "#b70600", "#a472d3", "#cdd321",
  "#dc68f9", "#10b251", "#f0c6ff", "#2f32bf", "#ef2882"
)
names(genus_colors) <- c("Other", rev(taxa_include))

ra_long_selected <- study_data %>%
  dplyr::select(Sample_ID, Participant_ID, Source_Material, Specimen_Storage_Time_Plot, dplyr::all_of(ra_cols))
  tidyr::pivot_longer(cols = dplyr::all_of(ra_cols), names_to = "Genera", values_to = "RA") %>%
  dplyr::mutate(Genera = sub("^RA_", "", Genera))

ra_other <- study_data %>%
  dplyr::select(Sample_ID, Participant_ID, Source_Material, Specimen_Storage_Time_Plot, dplyr::all_of(ra_cols))
  dplyr::mutate(RA = pmax(0, 1 - rowSums(dplyr::across(dplyr::all_of(ra_cols), na.rm = TRUE)))) %>%
  dplyr::transmute(Sample_ID, Participant_ID, Source_Material, Specimen_Storage_Time_Plot, Genera = "Other",
    RA)

ra_long <- dplyr::bind_rows(ra_long_selected, ra_other) %>%
  dplyr::mutate(
    Genera = factor(Genera, levels = c("Other", rev(taxa_include))),
    Participant_ID = factor(Participant_ID, levels = sort(unique(Participant_ID)))
  )

average_ra <- ra_long %>%
  dplyr::group_by(Specimen_Storage_Time_Plot, Source_Material, Genera) %>%
  dplyr::summarise(mean_RA = mean(RA, na.rm = TRUE), .groups = "drop")

p_figure1 <- ggplot(average_ra, aes(x = Specimen_Storage_Time_Plot, y = mean_RA, fill = Genera)) +
  geom_bar(stat = "identity", position = "fill") +
  scale_y_continuous(expand = c(0, 0), limits = c(0, 1)) +
  facet_grid(~ Source_Material, scales = "free_x", space = "free") +
  theme_bw() +
  labs(x = "Timepoint", y = "Relative Abundance") +
  scale_fill_manual(values = genus_colors, drop = FALSE) +

```

```

theme(axis.text.x = element_text(angle = 90, vjust = 0.3, size = 10), legend.text = element_text(size = 8))

png("Figure 1.png", width = 1200, height = 1500, res = 300)
print(p_figure1)
invisible(dev.off())

p_figure1

```

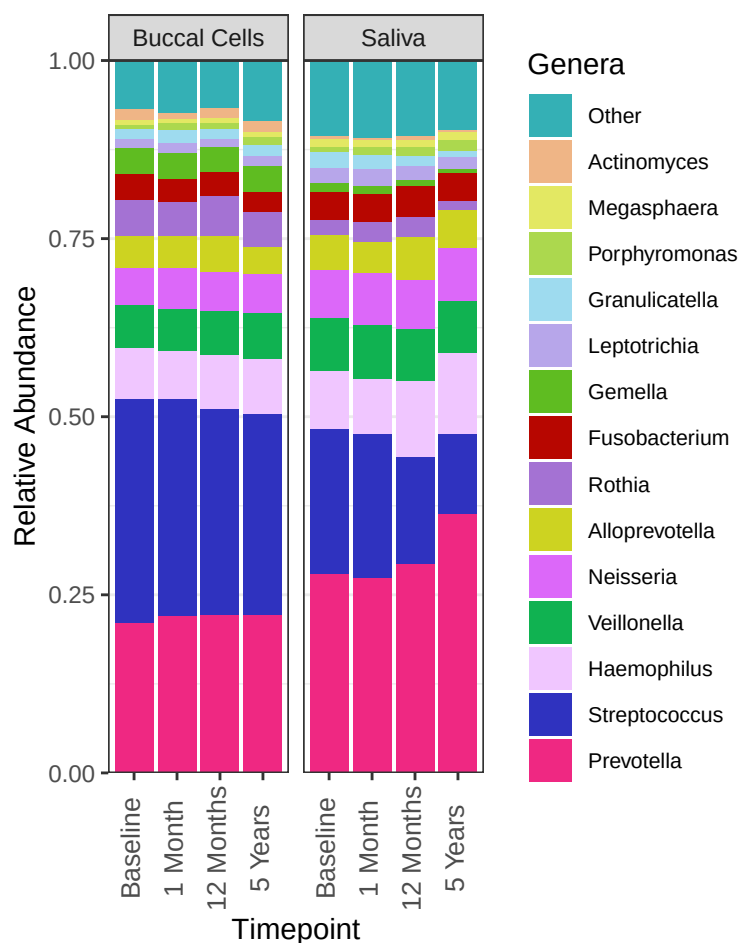


Figure 1: Average genus-level relative abundance for DNA extracted at baseline, after 1 month, 12 months, and 5 years from collection of the 14 genera with at least a relative abundance of 0.01 (i.e., 1%) grouped by sample collection type (i.e., oral wash/buccal cells or saliva).

5.2 Supplemental Figure 2

Supplemental Figure 2 shows genus-level relative abundance by participant for buccal cells.

```

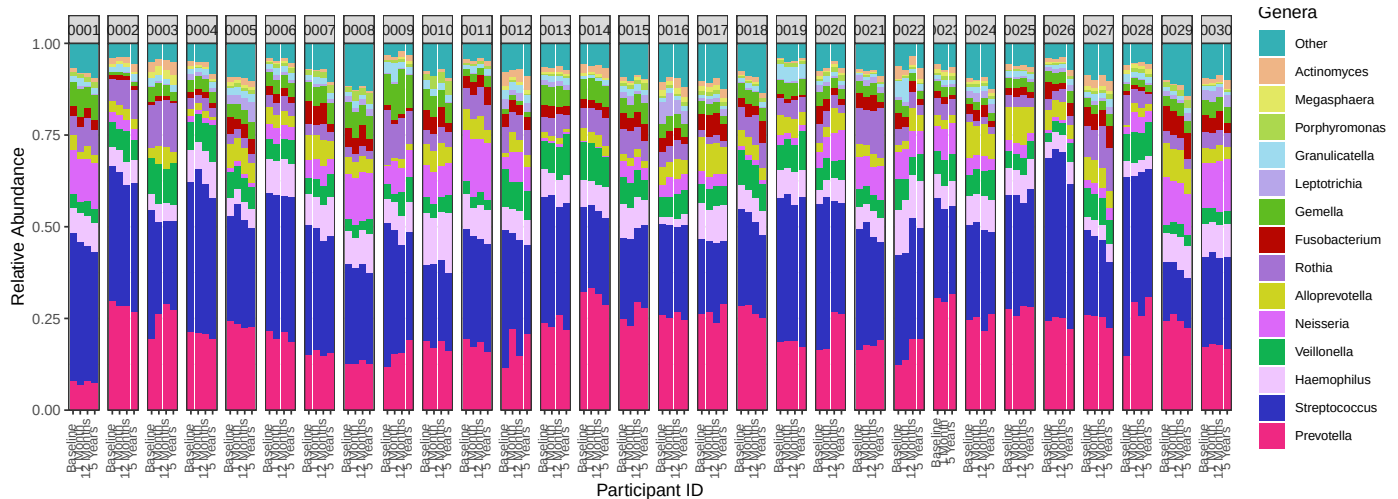
make_ra_by_participant_plot <- function(dat, axis_text_size) {
  ggplot(dat, aes(x = Specimen_Storage_Time_Plot, y = RA, fill = Genera)) +
    geom_bar(stat = "identity", position = "fill", width = 0.85) +
    scale_y_continuous(expand = c(0, 0), limits = c(0, 1)) +
    facet_grid(~ Participant_ID, scales = "free_x", space = "free") +
    theme_bw() +
    labs(x = "Participant ID", y = "Relative Abundance") +
    scale_fill_manual(values = genus_colors, drop = FALSE) +
    theme(axis.text.x = element_text(angle = 90, vjust = 0.3, size = axis_text_size), legend.text = element_t
}

```

```
p_supplemental2 <- make_ra_by_participant_plot(
  ra_long[ra_long$Source_Material == "Buccal Cells", , drop = FALSE],
  axis_text_size = 8
)

png("Supplemental Figure 2.png", width = 7000, height = 2500, res = 300)
print(p_supplemental2)
invisible(dev.off())

p_supplemental2
```



Supplemental Figure 2: Genus-level relative abundances from oral wash samples of the 14 most common genera grouped by participant ID at four timepoints (baseline, one month, 12 months, and 5 years).

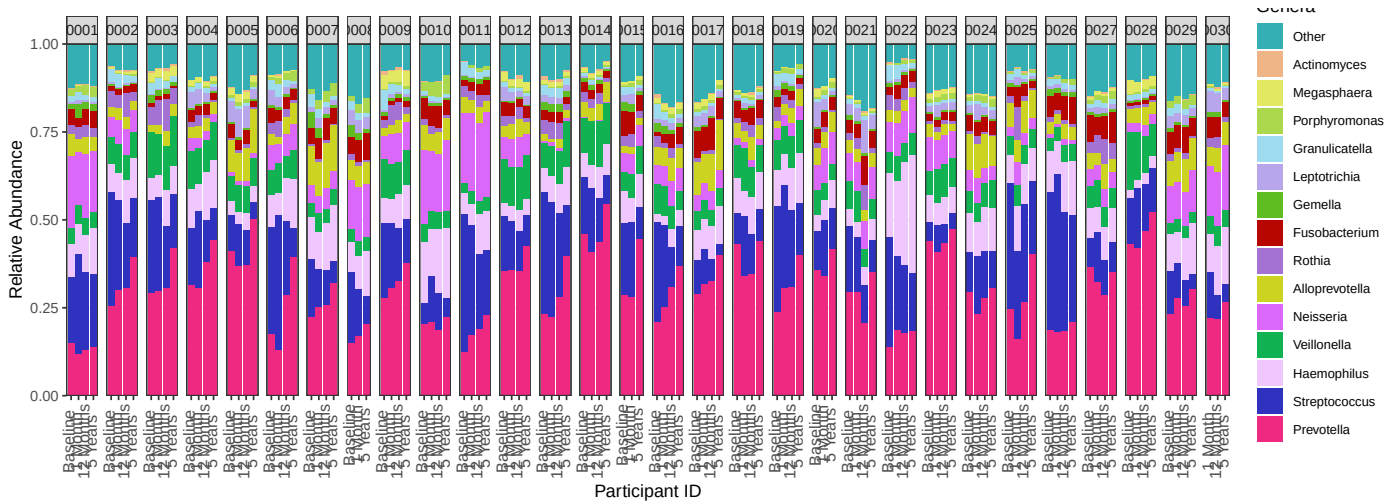
5.3 Supplemental Figure 3

Supplemental Figure 3 shows genus-level relative abundance by participant for saliva.

```
p_supplemental3 <- make_ra_by_participant_plot(
  ra_long[ra_long$Source_Material == "Saliva", , drop = FALSE],
  axis_text_size = 10
)

png("Supplemental Figure 3.png", width = 7000, height = 2500, res = 300)
print(p_supplemental3)
invisible(dev.off())

p_supplemental3
```



Supplemental Figure 3: Genus-level relative abundances from saliva samples of the 14 most common genera grouped by participant ID at four timepoints (baseline, one month, 12 months, and 5 years).

6. PERMANOVA / Adonis2 Analysis and Figure 2

The PERMANOVA / Adonis2 analysis is regenerated when `run_adnois` is TRUE. When the flag is FALSE, this section reads the copied Adonis2 result table.

```
if (run_adnois) {
  data_model <- study_data[, c("Source_Material", "Specimen_Storage_Time", "Subject_ID"), drop = FALSE]
  data_model$Source_Material <- factor(data_model$Source_Material)
  data_model$Specimen_Storage_Time <- factor(data_model$Specimen_Storage_Time, levels = c("Baseline", "Month1", "Month12", "Month5"))
  data_model$Subject_ID <- factor(data_model$Subject_ID, levels = levels(study_data$Subject_ID))

  adonis_results <- lapply(names(distance_files), function(distance_name) {
    dist_mat <- as.matrix(read.table(
      distance_files[[distance_name]],
      sep = "\t",
      header = TRUE,
      row.names = 1,
      check.names = FALSE
    ))
    storage.mode(dist_mat) <- "numeric"
    dist_mat <- dist_mat[rownames(data_model), rownames(data_model), drop = FALSE]

    vegan::adonis2(
      dist_mat ~ Source_Material + Specimen_Storage_Time + Subject_ID,
      data = data_model,
      by = "terms"
    )
  })
  names(adonis_results) <- names(distance_files)

  adonis2_r2 <- do.call(cbind, lapply(adonis_results, function(x) x$R2[1:4]))
  rownames(adonis2_r2) <- c("Source Material", "Specimen Storage Time", "Subject", "Residual")
  colnames(adonis2_r2) <- names(distance_files)
  write.csv(adonis2_r2, file = adonis2_file, quote = FALSE)
} else {
  message("Adonis2 calculation skipped; reading copied Adonis2 result table.")
  adonis2_r2 <- read.csv(adonis2_file, row.names = 1, check.names = FALSE, stringsAsFactors = FALSE)
}
```

```
format_table(adonis2_r2, digits = 9, caption = "PERMANOVA / Adonis2 R2 summary.")
```

PERMANOVA / Adonis2 R2 summary.

	Bray Curtis	Jaccard	Unweighted Unifrac	Generalized Unifrac	Weighted Unifrac
Source Material	0.08691012	0.01149667	0.01498059	0.16753418	0.16415040
Specimen Storage Time	0.01725046	0.01217652	0.01981475	0.03090131	0.03113986
Subject	0.79331465	0.76264796	0.70369957	0.62531339	0.62636543
Residual	0.10252477	0.21367884	0.26150509	0.17625112	0.17834431

```
adonis2_long <- data.frame(
  Term = rep(rownames(adonis2_r2), ncol(adonis2_r2)),
  Distance = rep(colnames(adonis2_r2), each = nrow(adonis2_r2)),
  R2 = as.vector(as.matrix(adonis2_r2)),
  check.names = FALSE
)
adonis2_long$Term <- factor(adonis2_long$Term, levels = c("Residual", "Subject", "Source Material", "Specimen Storage Time"))
adonis2_long$Distance <- factor(
  adonis2_long$Distance,
  levels = c("Bray Curtis", "Jaccard", "Unweighted Unifrac", "Generalized Unifrac", "Weighted Unifrac")
)

p_adonis2 <- ggplot(adonis2_long, aes(x = Distance, y = R2, fill = Term)) +
  geom_col(width = 0.8) +
  scale_y_continuous(limits = c(0, 1.01), breaks = seq(0, 1, 0.2), labels = scales::percent) +
  labs(x = "Beta diversity", y = expression("Explained variance (R^2*)"), fill = NULL) +
  theme_bw(base_size = 14) +
  theme(panel.grid.major.x = element_blank(), axis.text.x = element_text(angle = 45, hjust = 1))

png("Figure 2.png", width = 2000, height = 2000, res = 300)
print(p_adonis2)
invisible(dev.off())

p_adonis2
```

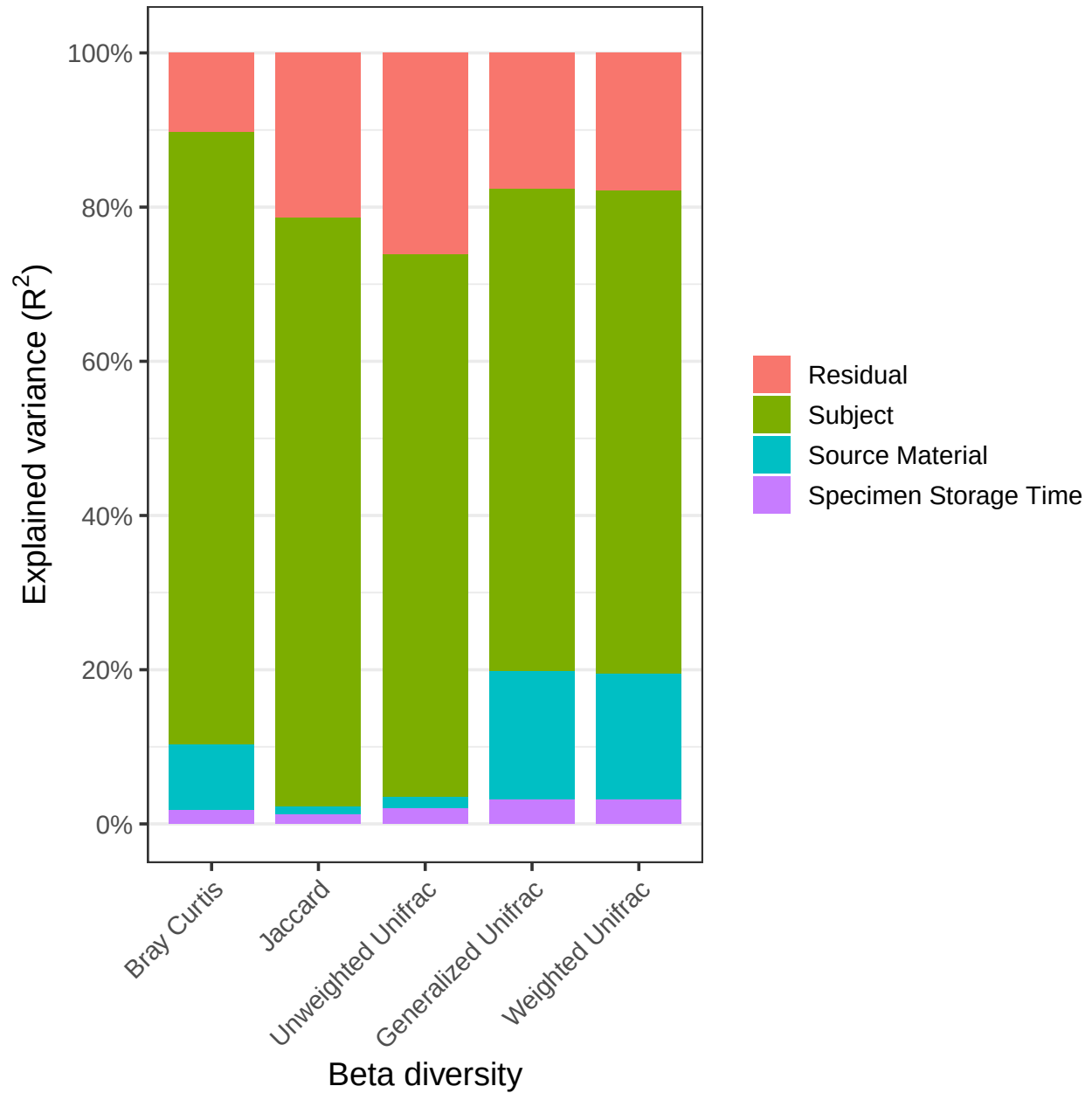


Figure 2: Percent variability explained in beta diversity (i.e., Bray Curtis, Jaccard, unweighted Unifrac, generalized Unifrac, and weighted Unifrac) by specimen storage time (i.e., time in the freezer), source material (i.e., oral wash or saliva), and subject.

7. ICC / 95% CI Calculation

The ICC and 95% CI calculation is regenerated when `run_icc_calculation` is TRUE. When the flag is FALSE, this section reads the copied ICC txt files.

```
comparison_labels <- c(Month1 = "1 Month", Month12 = "12 Months", Year5 = "5 Years")
comparison_times <- names(comparison_labels)
comparison_seed_index <- c(Month1 = 1L, Month12 = 4L, Year5 = 5L)
icc_stats <- c(
  "ICC_adjusted",
  "ICC_adjusted_CI_lower",
  "ICC_adjusted_CI_upper",
```

```

"ICC_unadjusted",
"ICC_unadjusted_CI_lower",
"ICC_unadjusted_CI_upper"
)
icc_wide_cols <- as.vector(t(outer(unname(comparison_labels), c("ICC", "LL", "UL"), paste, sep = "_")))

read_icc_wide <- function(path) {
  out <- read.table(path, sep = "\t", row.names = 1, check.names = FALSE, header = FALSE, na.strings = "NA")
  if (ncol(out) != length(icc_wide_cols)) {
    stop("Expected ", length(icc_wide_cols), " ICC columns in ", path, ", found ", ncol(out), ".")
  }
  colnames(out) <- icc_wide_cols
  out
}

make_long_icc <- function(wide, source_material) {
  wide <- tibble::rownames_to_column(as.data.frame(wide, check.names = FALSE), "Outcome")
  wide %>%
    tidyr::pivot_longer(!Outcome, names_to = "Timepoint", values_to = "Value") %>%
    dplyr::mutate(
      Specimen_Storage_Time_Comparison = sub("_.*", "", Timepoint),
      Statistic = sub(".*_", "", Timepoint)
    ) %>%
    tidyr::pivot_wider(
      id_cols = c(Outcome, Specimen_Storage_Time_Comparison),
      names_from = Statistic,
      values_from = Value
    ) %>%
    dplyr::rename(
      ICC_adjusted = ICC,
      ICC_adjusted_CI_lower = LL,
      ICC_adjusted_CI_upper = UL
    ) %>%
    dplyr::mutate(Source_Material = source_material) %>%
    dplyr::select(Source_Material, Outcome, Specimen_Storage_Time_Comparison, dplyr::everything())
}

if (run_icc_calculation) {
  library(lme4)
  library(performance)

  calc_icc <- function(dat, outcome) {
    dat <- dat[!is.na(dat[[outcome]]), , drop = FALSE]
    if (nrow(dat) == 0L || length(unique(dat$Specimen_Storage_Time)) < 2L) {
      return(rep(NA_real_, length(icc_stats)))
    }

    model <- try(
      lme4::lmer(
        stats::as.formula(paste0(outcome, " ~ Specimen_Storage_Time + (1 | Subject_ID)")),
        data = dat
      ),
      silent = TRUE
    )
    if (inherits(model, "try-error")) {
      return(rep(NA_real_, length(icc_stats)))
    }
  }
}

```



```

variance_components <- lme4::VarCorr(model)
sigma_between <- attr(variance_components$Subject_ID, "stddev")
sigma_error <- attr(variance_components, "sc")
fallback_icc <- sigma_between^2 / (sigma_between^2 + sigma_error^2)

icc_result <- try(
  performance::icc(model, tolerance = 1e-7, ci = TRUE, iterations = icc_iterations),
  silent = TRUE
)
out <- rep(NA_real_, length(icc_stats))
names(out) <- icc_stats

if (inherits(icc_result, "try-error") || any(is.na(icc_result))) {
  out["ICC_adjusted"] <- fallback_icc
  return(out)
}

out[] <- c(icc_result$ICC_adjusted, icc_result$ICC_unadjusted)
out
}

run_source_icc <- function(source_material) {
  source_data <- study_data[study_data$Source_Material == source_material, , drop = FALSE]
  source_data$Subject_ID <- droplevels(source_data$Subject_ID)
  icc_array <- array(
    NA_real_,
    dim = c(length(outcome_vars), length(comparison_times), length(icc_stats)),
    dimnames = list(outcome_vars, comparison_times, icc_stats)
  )

  for (j in seq_along(comparison_times)) {
    comparison <- comparison_times[[j]]
    comparison_data <- source_data[source_data$Specimen_Storage_Time %in% c("Baseline", comparison), , drop = FALSE]
    comparison_data$Specimen_Storage_Time <- droplevels(comparison_data$Specimen_Storage_Time)
    comparison_data$Subject_ID <- droplevels(comparison_data$Subject_ID)

    for (i in seq_along(outcome_vars)) {
      set.seed(unname(comparison_seed_index[comparison]) * 10000 + i)
      icc_array[i, j, ] <- calc_icc(comparison_data, outcome_vars[[i]])
    }
  }

  matrix(
    aperm(icc_array[, , 1:3, drop = FALSE], c(1, 3, 2)),
    nrow = dim(icc_array)[1],
    ncol = length(comparison_times) * 3,
    dimnames = list(outcome_vars, icc_wide_cols)
  )
}

wide_buccal <- run_source_icc("Buccal Cells")
wide_saliva <- run_source_icc("Saliva")

write.table(wide_buccal, file = icc_buccal_file, sep = "\t", quote = FALSE, row.names = TRUE, col.names = TRUE)
write.table(wide_saliva, file = icc_saliva_file, sep = "\t", quote = FALSE, row.names = TRUE, col.names = TRUE)
} else {
  message("ICC calculation skipped; reading copied ICC txt files.")
}

```

```

wide_buccal <- read_icc_wide(icc_buccal_file)
wide_saliva <- read_icc_wide(icc_saliva_file)

long_icc <- dplyr::bind_rows(
  make_long_icc(wide_buccal, "Buccal Cells"),
  make_long_icc(wide_saliva, "Saliva")
)

write_csv(long_icc, file = "result_ICC_Storage_Time_long.csv", row.names = FALSE, quote = FALSE, na = "NA")

format_table(wide_buccal, digits = 8, caption = "Buccal cell ICC and 95% CI summary.", wide = TRUE)

```

Buccal cell ICC and 95% CI summary.

	1 Month_ICC	1 Month_LL	1 Month_UL	12 Months_ICC	12 Months_LL	12 Months_UL	5 Years_ICC	5 Years_LL	5 Years_UL
observed_features	0.9424550	0.8652923	0.9766058	0.9134392	0.8057105	0.9658739	0.9582740	0.9033882	0.9821441
shannon_entropy	0.9464749	0.8971691	0.9774425	0.9383775	0.8624699	0.9746484	0.9396001	0.8776560	0.9755005
pielou_evenness	0.8782330	0.6917954	0.9413356	0.9196058	0.8072891	0.9585105	0.8922272	0.7421639	0.9573789
faith_pd	0.9283065	0.8207346	0.9666910	0.8731184	0.6784823	0.9375752	0.9384108	0.8249606	0.9752847
bray_curtis_PC1	0.9870975	0.9684023	0.9960337	0.9890754	0.9477001	0.9964139	0.9725401	0.9325175	0.9902805
bray_curtis_PC2	0.9836433	0.9558058	0.9923173	0.9762522	0.9145298	0.9911660	0.9774916	0.9453470	0.9912370
jaccard_PC1	0.9617508	0.9224170	0.9845865	0.9840980	0.9619184	0.9936866	0.9687398	0.9292751	0.9916320
jaccard_PC2	0.9747783	0.9439168	0.9902006	0.9869555	0.9647517	0.9951471	0.9821374	0.9664746	0.9938948
unweighted_unifrac_PC1	0.9569411	0.8748717	0.9766703	0.9359204	0.8576034	0.9635105	0.9550382	0.9193768	0.9876403
unweighted_unifrac_PC2	0.9470998	0.8588408	0.9776840	0.9346529	0.8628499	0.9686547	0.9569803	0.8832998	0.9825035
generalized_unifrac_PC1	0.8765056	0.7210651	0.9634688	0.8734403	0.6274166	0.9531985	0.7618555	0.4308473	0.9084150
generalized_unifrac_PC2	0.9283871	0.8349337	0.9671047	0.9211517	0.8044059	0.9695619	0.8920794	0.7784813	0.9447878
weighted_unifrac_PC1	0.8500545	0.4264001	0.9292113	0.8479905	0.6475542	0.9394355	0.7280357	0.4024289	0.8896976
weighted_unifrac_PC2	0.9483581	0.8711448	0.9787322	0.9535107	0.8899481	0.9835373	0.9450527	0.8354008	0.9760849
RA_Neisseria	0.9414598	0.8549639	0.9857487	0.9436541	0.8708946	0.9741941	0.9532298	0.8868094	0.9751500
RA_Megasphaera	0.9064496	0.7890623	0.9736662	0.9230252	0.8529405	0.9741401	0.8389603	0.6334855	0.9358640
RA_Prevotella	0.8215524	0.6025329	0.9219165	0.8147136	0.5777331	0.9288852	0.6760925	0.4179501	0.8403647
RA_Streptococcus	0.8577044	0.5740468	0.9340450	0.8737452	0.7291473	0.9546694	0.8291484	0.6268318	0.9290814
RA_Fusobacterium	0.8350626	0.5436683	0.9358423	0.8307022	0.4600211	0.9423736	0.5862521	0.2526595	0.7927078
RA_Leptotrichia	0.9368029	0.8393923	0.9745939	0.8607509	0.6064026	0.9474497	0.7825513	0.5588850	0.8710467
RA_Alloprevotella	0.8756993	0.7178051	0.9401026	0.8259095	0.6608720	0.9271912	0.7755918	0.5762411	0.9155222
RA_Porphyrionas	0.7920000	0.5684637	0.9080955	0.8776896	0.6465820	0.9592250	0.7308929	0.4986937	0.8810041
RA_Rothia	0.8645748	0.7248786	0.9386658	0.8651408	0.6648123	0.9385364	0.7990011	0.5153741	0.9261510
RA_Veillonella	0.8846148	0.7386284	0.9588225	0.8532964	0.5795599	0.9385367	0.6959559	0.3208978	0.8620198
RA_Gemella	0.9698059	0.9227188	0.9877481	0.8811279	0.7274081	0.9684177	0.8916361	0.7014080	0.9671878
RA_Actinomyces	0.7538538	0.3979770	0.8770458	0.9211316	0.8372801	0.9633245	0.9150815	0.8214805	0.9672100
RA_Haemophilus	0.9294809	0.7992472	0.9816251	0.9429366	0.8481797	0.9748935	0.9310940	0.8651683	0.9759389
RA_Granulicatella	0.8330970	0.6752019	0.9451160	0.5770253	0.1079849	0.7899875	0.5673884	0.1734050	0.8161022
RA_clr_Neisseria	0.9636957	0.8902674	0.9887403	0.9691970	0.9340483	0.9886942	0.9674857	0.9236148	0.9874662
RA_clr_Megasphaera	0.9703476	0.9143228	0.9900590	0.9565354	0.8725419	0.9877498	0.9200277	0.8122738	0.9705804
RA_clr_Prevotella	0.8311246	0.5972252	0.9178820	0.8340195	0.5200505	0.9173050	0.6980112	0.4151310	0.8359461
RA_clr_Streptococcus	0.9045025	0.7914413	0.9629463	0.9076460	0.7334010	0.9631688	0.8831861	0.7471397	0.9553265
RA_clr_Fusobacterium	0.8711306	0.6671516	0.9649126	0.8665557	0.6941342	0.9418936	0.7121901	0.3333291	0.9084057
RA_clr_Leptotrichia	0.8874087	0.6574613	0.9520454	0.8664257	0.7359493	0.9382429	0.8253719	0.6763214	0.9229354
RA_clr_Alloprevotella	0.7575808	0.4995748	0.8788530	0.6381183	0.1872593	0.8291546	0.6259145	0.1778311	0.8312404
RA_clr_Porphyrionas	0.8837533	0.7859687	0.9610663	0.9022036	0.7934146	0.9674030	0.8631342	0.6270725	0.9363875
RA_clr_Rothia	0.9079557	0.7759534	0.9638839	0.8985141	0.7199159	0.9566347	0.8424129	0.5840371	0.9334507
RA_clr_Veillonella	0.8769247	0.7195471	0.9553846	0.8504547	0.6678808	0.9293357	0.7479816	0.3815339	0.8950296
RA_clr_Gemella	0.9718585	0.9308751	0.9895843	0.9393044	0.8043160	0.9810651	0.8681873	0.6642945	0.9415466
RA_clr_Actinomyces	0.8297166	0.6026759	0.9261393	0.9035564	0.8105711	0.9676149	0.8775728	0.7282573	0.9572189
RA_clr_Haemophilus	0.9391838	0.8772684	0.9737115	0.9372612	0.8527474	0.9729069	0.9133000	0.7868005	0.9653809
RA_clr_Granulicatella	0.8481415	0.7019805	0.9465316	0.6937062	0.4442669	0.8670169	0.6384106	0.3032304	0.8529105

```

format_table(wide_saliva, digits = 8, caption = "Saliva ICC and 95% CI summary.", wide = TRUE)

```

Saliva ICC and 95% CI summary.

	1 Month_ICC	1 Month_LL	1 Month_UL	12 Months_ICC	12 Months_LL	12 Months_UL	5 Years_ICC	5 Years_LL	5 Years_UL
observed_features	0.9521599	0.89511929	0.9842084	0.9290513	0.81947232	0.9668436	0.8944426	0.73288233	0.9597320
shannon_entropy	0.9414987	0.88314078	0.9754663	0.9378734	0.84011511	0.9703165	0.8968181	0.72017433	0.9527335
pielou_evenness	0.8829683	0.73269869	0.9456578	0.8838489	0.69295319	0.9479237	0.8628336	0.70356298	0.9511704
faith_pd	0.9289784	0.82252920	0.9723383	0.9240544	0.79346761	0.9745278	0.8806440	0.75353077	0.9615882
bray_curtis_PC1	0.9815020	0.95783288	0.9916995	0.9802182	0.95751469	0.9910468	0.9797817	0.94925733	0.9919977
bray_curtis_PC2	0.9822624	0.95306188	0.9944313	0.9840511	0.96946555	0.9934347	0.9595774	0.91699007	0.9825761
jaccard_PC1	0.9689299	0.91456011	0.9899703	0.9774398	0.94700122	0.9917041	0.9658541	0.90826177	0.9907179
jaccard_PC2	0.9885684	0.96540808	0.9960054	0.9950091	0.98745808	0.9980099	0.9874573	0.96291337	0.9950901
unweighted_unifrac_PC1	0.9468896	0.83081072	0.9799207	0.9413981	0.82994208	0.9766118	0.9533388	0.90075324	0.9843411
unweighted_unifrac_PC2	0.9590544	0.89055312	0.9862154	0.9543779	0.82645924	0.9834514	0.9371964	0.84377895	0.9720804
generalized_unifrac_PC1	0.9302641	0.86406161	0.9720245	0.8969094	0.75401839	0.9570665	0.8941150	0.64608023	0.9536529
generalized_unifrac_PC2	0.9477372	0.86052240	0.9791931	0.9611473	0.90540700	0.9859081	0.9054815	0.77105898	0.9692624
weighted_unifrac_PC1	0.9205524	0.74948595	0.9692220	0.8777941	0.73883012	0.9424115	0.8553605	0.67724240	0.9206500
weighted_unifrac_PC2	0.9488247	0.85488136	0.9860151	0.9752907	0.94132585	0.9927196	0.9495196	0.87887255	0.9782965
RA_Neisseria	0.9489633	0.81968053	0.9792706	0.9663370	0.91900880	0.9906386	0.9749334	0.92633485	0.9892763
RA_Megasphaera	0.9345378	0.84283705	0.9699492	0.9492214	0.89490418	0.9806737	0.9545696	0.90621833	0.9843680
RA_Prevotella	0.8925816	0.75314945	0.9455118	0.8334459	0.58816131	0.9425539	0.8221504	0.67939102	0.9273793
RA_Streptococcus	0.8228761	0.59067455	0.9324576	0.7664771	0.50242358	0.9351479	0.6580715	0.31857021	0.8709009
RA_Fusobacterium	0.8022164	0.53670842	0.9350625	0.8138554	0.62789015	0.9246490	0.7170942	0.35216968	0.9222818
RA_Leptotrichia	0.8078080	0.58659798	0.9117160	0.7099042	0.22401279	0.8593990	0.5409073	0.18399998	0.7620167
RA_Alloprevotella	0.9305747	0.78094180	0.9715613	0.8930396	0.81289352	0.9544567	0.8462746	0.67565539	0.9282366
RA_Porphyrromonas	0.8767160	0.73773817	0.9461798	0.8592654	0.68470880	0.9547198	0.7328710	0.51468509	0.8790086
RA_Rothia	0.4549766	0.02427406	0.7281972	0.5850451	0.10593497	0.8092815	0.6698979	0.16392116	0.8513551
RA_Veillonella	0.8926137	0.79426450	0.9589123	0.8193435	0.50569802	0.9142296	0.7452347	0.49175570	0.9221307
RA_Gemella	0.7112985	0.41257782	0.8827430	0.5067845	0.16939326	0.7906407	0.3881560	0.01043444	0.7781632
RA_Actinomyces	0.3855902	0.03350256	0.7692583	0.5336647	0.19823579	0.7812783	0.4428515	0.06018019	0.7289844
RA_Haemophilus	0.8777438	0.70311189	0.9534203	0.8669390	0.47573641	0.9512548	0.8293690	0.65886152	0.9202969
RA_Granulicatella	0.7446270	0.44538006	0.9309606	0.4522759	0.01069612	0.7760993	0.3653924	0.01087041	0.7467642
RA_clr_Neisseria	0.9760263	0.94971308	0.9932368	0.9849881	0.95352127	0.9924785	0.9862663	0.96717165	0.9952104
RA_clr_Megasphaera	0.9851547	0.95828082	0.9958263	0.9901738	0.97377055	0.9961724	0.9838925	0.95339599	0.9944637
RA_clr_Prevotella	0.8714443	0.72650692	0.9483214	0.8198682	0.53365964	0.9229774	0.8247803	0.63887507	0.9557638
RA_clr_Streptococcus	0.7994899	0.61224393	0.9080739	0.8139847	0.54648693	0.9249206	0.7641630	0.48671390	0.8925286
RA_clr_Fusobacterium	0.8375554	0.61327661	0.9367622	0.8142900	0.53593501	0.9346318	0.6799820	0.22501861	0.8943087
RA_clr_Leptotrichia	0.8765376	0.70273587	0.9453073	0.8915132	0.68681257	0.9620764	0.7928761	0.51494962	0.9191200
RA_clr_Alloprevotella	0.8836244	0.77257344	0.9473038	0.8310786	0.58077083	0.9340743	0.8130413	0.51691212	0.9286624
RA_clr_Porphyrromonas	0.9232848	0.83681665	0.9761593	0.8971022	0.77352030	0.9587141	0.8883028	0.74561952	0.9508938
RA_clr_Rothia	0.6853938	0.35790090	0.8255699	0.7706152	0.51189951	0.9011334	0.7718038	0.42702752	0.9056050
RA_clr_Veillonella	0.9192863	0.82052987	0.9702161	0.8817310	0.68370711	0.9475701	0.8133208	0.58659857	0.9111688
RA_clr_Gemella	0.8264694	0.53994996	0.9415403	0.7511882	0.34316743	0.9075129	0.6962166	0.32629787	0.8579264
RA_clr_Actinomyces	0.4229484	0.00176507	0.7344256	0.5387084	0.17186231	0.7957241	0.5685253	0.16853409	0.8176844
RA_clr_Haemophilus	0.9213971	0.82216607	0.9646019	0.9052995	0.78255910	0.9633068	0.9304145	0.85667744	0.9712008
RA_clr_Granulicatella	0.8341983	0.65868755	0.9339519	0.7185652	0.39219748	0.9192858	0.6834410	0.36180757	0.9028423

7.1 Figure 3

Figure 3 shows ICC estimates and 95% CIs for alpha- and beta-diversity metrics.

```
pretty_alpha_beta <- c(
  "Observed ASVs",
  "Shannon Index",
  "Pielou's evenness",
  "Faith's phylogenetic diversity",
  "Bray Curtis (PC1)",
  "Bray Curtis (PC2)",
  "Jaccard (PC1)",
  "Jaccard (PC2)",
  "Unweighted Unifrac (PC1)",
  "Unweighted Unifrac (PC2)",
  "Generalized Unifrac (PC1)",
  "Generalized Unifrac (PC2)",
  "Weighted Unifrac (PC1)",
  "Weighted Unifrac (PC2)"
)

rename_icc_rows <- function(x) {
  rownames(x)[seq_along(pretty_alpha_beta)] <- pretty_alpha_beta
  rownames(x)[15:28] <- sub("RA_", "", rownames(x)[15:28])
  tibble::rownames_to_column(as.data.frame(x, check.names = FALSE), "Index")
}

make_icc_plot_data <- function(wide_data, row_index, collection) {
  wide_data[row_index, , drop = FALSE] %>%
    tidyr::pivot_longer(!Index, names_to = "Timepoint", values_to = "ICC") %>%
    dplyr::mutate(
      Time = sub("_.*", "", Timepoint),

```

```

    Timepoint = sub(".*_", "", Timepoint)
  ) %>%
  tidyr::pivot_wider(names_from = Timepoint, values_from = ICC) %>%
  dplyr::mutate(Collection = collection)
}

icc_buccal_wide <- rename_icc_rows(wide_buccal)
icc_saliva_wide <- rename_icc_rows(wide_saliva)

data_plot_alpha_beta <- dplyr::bind_rows(
  make_icc_plot_data(icc_buccal_wide, 1:14, "Buccal Cells"),
  make_icc_plot_data(icc_saliva_wide, 1:14, "Saliva")
)

p_figure3 <- ggplot(
  data = data_plot_alpha_beta,
  aes(x = forcats::fct_inorder(Index), y = ICC, ymin = LL, ymax = UL, colour = Collection)
) +
  geom_pointrange(position = position_dodge(width = 0.6), aes(shape = Time)) +
  scale_color_grey() +
  theme_classic() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  ylim(0, 1) +
  xlab(label = "Alpha and Beta Diversity Metrics") +
  guides(color = guide_legend(title = "Collection Type"))

png("Figure 3.png", width = 3200, height = 1400, res = 300)
print(p_figure3)
invisible(dev.off())

p_figure3

```

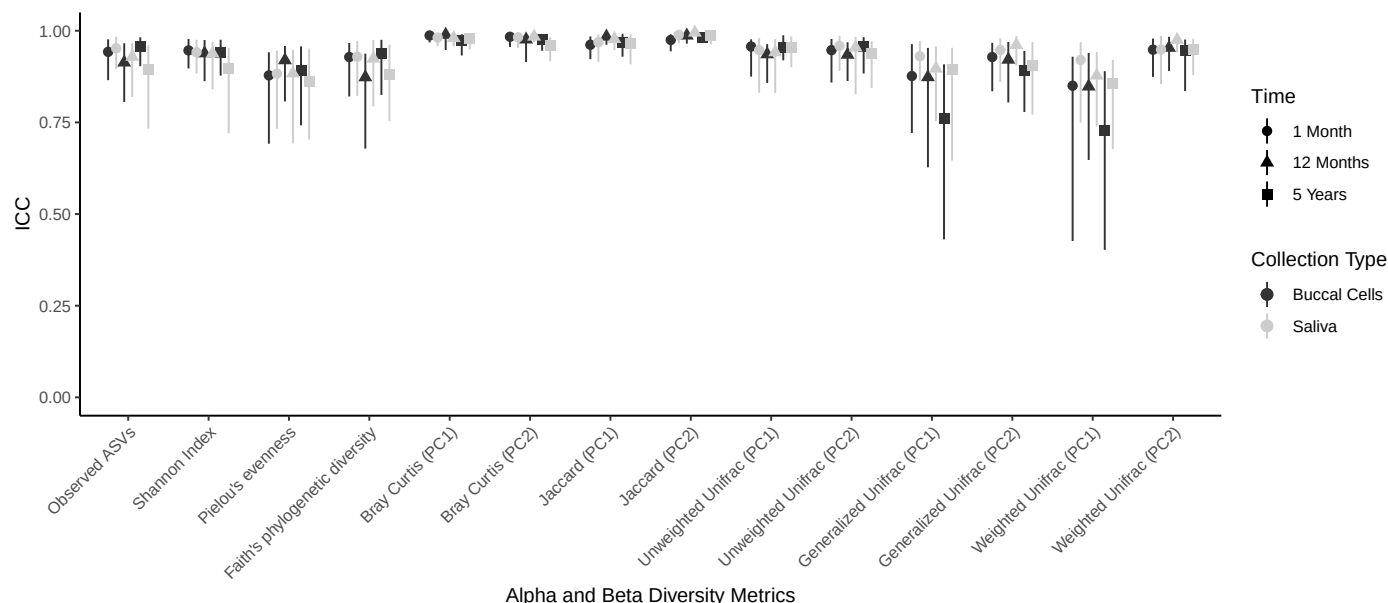


Figure 3: Interclass correlation coefficients and 95% confidence intervals for alpha and beta diversity metrics from oral wash/buccal cells and saliva samples with DNA extraction at 1 month, 12 months, and 5 years compared to the baseline DNA extraction. ICC = interclass correlation coefficient; ASV = amplicon sequence variant; PC = principal coordinate analysis vector.

7.2 Figure 4

Figure 4 shows ICC estimates and 95% CIs for the selected genera.

```
data_plot_taxa <- dplyr::bind_rows(
  make_icc_plot_data(icc_buccal_wide, 15:28, "Buccal Cells"),
  make_icc_plot_data(icc_saliva_wide, 15:28, "Saliva")
)

p_figure4 <- ggplot(
  data = data_plot_taxa,
  aes(x = forcats::fct_inorder(Index), y = ICC, ymin = LL, ymax = UL, colour = Collection)
) +
  geom_pointrange(position = position_dodge(width = 0.6), aes(shape = Time)) +
  scale_color_grey() +
  theme_classic() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  ylim(0, 1) +
  xlab(label = "Genera") +
  guides(color = guide_legend(title = "Collection Type"))

png("Figure 4.png", width = 3200, height = 1400, res = 300)
print(p_figure4)
invisible(dev.off())

p_figure4
```

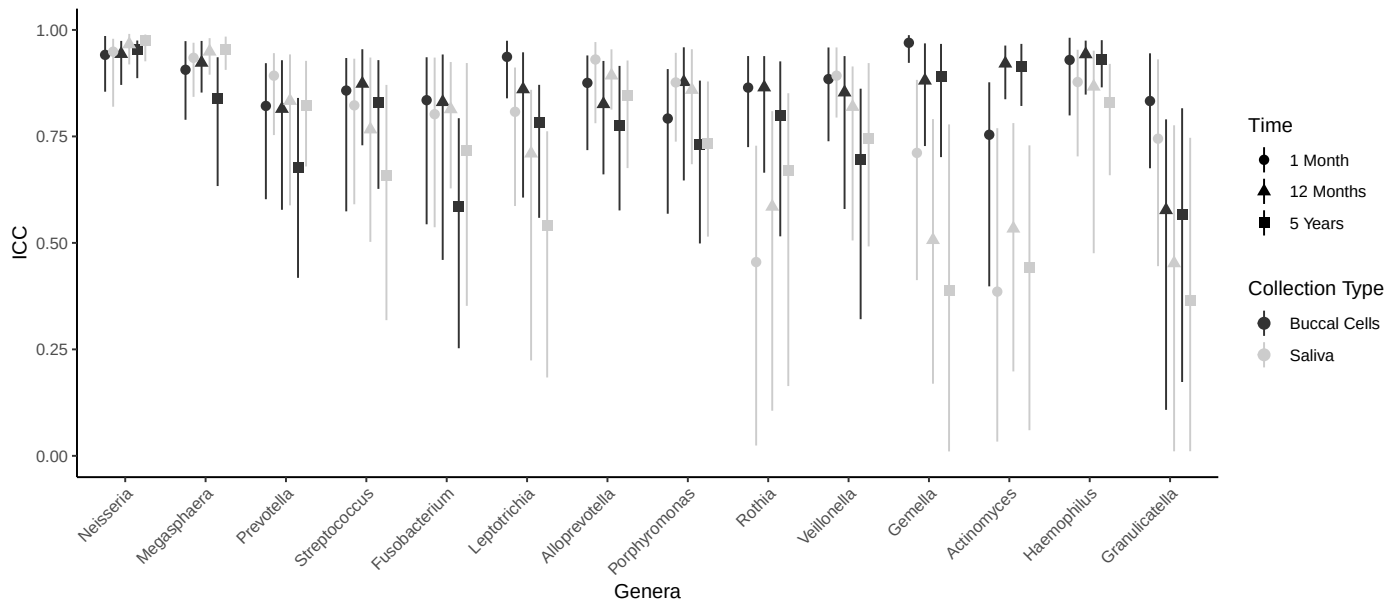


Figure 4: Interclass correlation coefficients and 95% confidence intervals for genus-level relative abundances from oral wash/buccal cells and saliva samples with DNA extraction at 1 month, 12 months, and 5 years compared to the baseline DNA extraction. The genera presented represent the 14 taxa that met the average relative abundance criteria of greater than 0.01 (i.e., 1%). ICC = interclass correlation coefficient

8. Supplemental Figure 4

Supplemental Figure 4 shows each participant's alpha-diversity trajectory across storage time, with the mean trajectory overlaid in red.

```

alpha_plot_data <- study_data %>%
  dplyr::select(
    Participant_ID,
    Source_Material,
    Specimen_Storage_Time_Plot,
    observed_features,
    shannon_entropy,
    pielou_evenness,
    faith_pd
  )

make_alpha_subject_plot <- function(dat, y_col, y_label, title = NULL) {
  ggplot(dat, aes(x = Specimen_Storage_Time_Plot, y = .data[[y_col]], group = Participant_ID)) +
    geom_line(color = "grey50", alpha = 0.5, linewidth = 0.7) +
    geom_point(color = "grey30", size = 2) +
    stat_summary(aes(group = 1), fun = mean, geom = "line", color = rgb(1, 0, 0, 0.2), linewidth = 1.5) +
    stat_summary(aes(group = 1), fun = mean, geom = "point", color = rgb(1, 0, 0, 0.2), size = 3) +
    labs(title = title, x = "Specimen Storage Time", y = y_label)
}

buccal_alpha <- alpha_plot_data[alpha_plot_data$Source_Material == "Buccal Cells", , drop = FALSE]
saliva_alpha <- alpha_plot_data[alpha_plot_data$Source_Material == "Saliva", , drop = FALSE]

p_alpha_buccal_observed <- make_alpha_subject_plot(buccal_alpha, "observed_features", "Observed ASVs", "Buccal"
  theme(plot.title = element_text(hjust = 0, size = 15, face = "bold", color = "black"))
p_alpha_buccal_shannon <- make_alpha_subject_plot(buccal_alpha, "shannon_entropy", "Shannon Index")
p_alpha_buccal_pielou <- make_alpha_subject_plot(buccal_alpha, "pielou_evenness", "Pielou's evenness")
p_alpha_buccal_faith <- make_alpha_subject_plot(buccal_alpha, "faith_pd", "Faith's phylogenetic diversity")

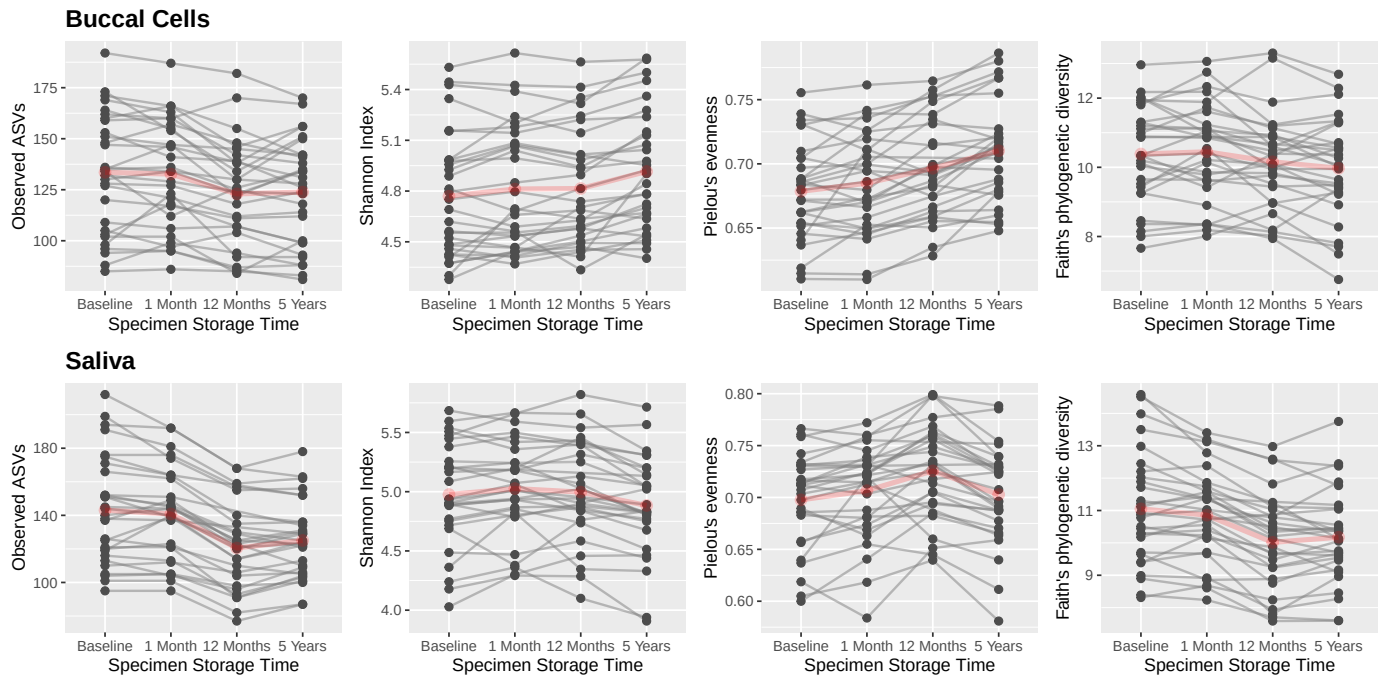
p_alpha_saliva_observed <- make_alpha_subject_plot(saliva_alpha, "observed_features", "Observed ASVs", "Saliva"
  theme(plot.title = element_text(hjust = 0, size = 15, face = "bold", color = "black"))
p_alpha_saliva_shannon <- make_alpha_subject_plot(saliva_alpha, "shannon_entropy", "Shannon Index")
p_alpha_saliva_pielou <- make_alpha_subject_plot(saliva_alpha, "pielou_evenness", "Pielou's evenness")
p_alpha_saliva_faith <- make_alpha_subject_plot(saliva_alpha, "faith_pd", "Faith's phylogenetic diversity")

p_alpha_vs_time <- (
  p_alpha_buccal_observed | p_alpha_buccal_shannon | p_alpha_buccal_pielou | p_alpha_buccal_faith
) / (
  p_alpha_saliva_observed | p_alpha_saliva_shannon | p_alpha_saliva_pielou | p_alpha_saliva_faith
)

png("Supplemental Figure 4.png", width = 5000, height = 2500, res = 300)
print(p_alpha_vs_time)
invisible(dev.off())

p_alpha_vs_time

```



Supplemental Figure 4: Individual trends in alpha diversity estimated from oral wash/buccal cell samples and saliva samples extracted at four timepoints (baseline, one month, 12 months, and 5 years). ASV = amplicon sequence variant.

Generated Output Files

The table below lists the files generated by this one-file workflow.

```
generated_outputs <- c(
  "Supplemental Figure 1.pdf",
  "Supplemental Figure 1.png",
  "Figure 1.png",
  "Supplemental Figure 2.png",
  "Supplemental Figure 3.png",
  "result_adonis2.csv",
  "Figure 2.png",
  "result_ICC_Storage_Time_long.csv",
  "Figure 3.png",
  "Figure 4.png",
  "Supplemental Figure 4.png"
)

output_summary <- data.frame(
  File = basename(generated_outputs),
  Exists = file.exists(generated_outputs),
  Bytes = file.info(generated_outputs)$size,
  check.names = FALSE
)

format_table(output_summary, caption = "Generated files from the one-file workflow.")
```

Generated files from the one-file workflow.

File	Exists	Bytes
Supplemental Figure 1.pdf	TRUE	87257
Supplemental Figure 1.png	TRUE	256157
Figure 1.png	TRUE	40673
Supplemental Figure 2.png	TRUE	291170
Supplemental Figure 3.png	TRUE	277912
result_adonis2.csv	TRUE	504
Figure 2.png	TRUE	55557
result_ICC_Storage_Time_long.csv	TRUE	22974
Figure 3.png	TRUE	58659
Figure 4.png	TRUE	63631
Supplemental Figure 4.png	TRUE	347222